

Md Mohai Minul Islam

Chandranagar, Mymensingh 2280, Bangladesh
mohaiminuladil.islam@gmail.com — +880-1703712056
LinkedIn Profile

Research Interests

Machine Learning, Artificial Intelligence

Education

Khulna University of Engineering & Technology (KUET) Bangladesh
B.Sc. in Electrical and Electronic Engineering 2013 – 2017
CGPA: 3.05 / 4.00

Research Experience

Undergraduate Thesis: Electronic Bandstructure of Hollow GaAs Nanowires
Supervisor: [Dr. Ashraful Ghani Bhuiyan] 2016 – 2017

- Developed computational models using tight-binding methods to analyze electronic band structure of GaAs nanowires
- Investigated the effect of inner and outer radius variation on bandgap modulation
- Implemented simulations in MATLAB to study quantum confinement effects in nanoscale systems

Projects

Free Space Optical Communication System Using LASER

- Built a prototype system for transmitting digitized audio signals via free-space optical link
- Designed ADC-DAC conversion pipeline and evaluated signal integrity under noise conditions

CMOS Inverter Design and Simulation

- Simulated static and dynamic characteristics using SPICE
- Analyzed switching behavior, power consumption, and noise margins

Technical Skills

Programming: MATLAB, Python, C/C++, JavaScript

Simulation Tools: COMSOL Multiphysics, SPICE

Areas: Semiconductor Modeling, Numerical Simulation, Signal Processing

Teaching Experience

Founder and Instructor, Apex Mathematics & Physics Program 2021 – Present

- Founded and led an academic program preparing students for competitive university admissions
- Taught advanced mathematics and physics to over 200 students
- Promoted problem-solving culture through Math Olympiad and Science Olympiad training

Mathematics Instructor, Omeca Coaching Centre 2020 – 2021

- Delivered structured lessons in mathematics to large student cohorts

Industry Experience

Engineer, Samuda Chemical Complex Ltd.

2018 – 2020

- Managed electrical systems including transformers, rectifiers, and industrial motors
- Oversaw power distribution and ensured operational reliability of plant systems

Intern, Samuda Chemical Complex Ltd.

2017 – 2018

- Assisted in monitoring and maintenance of electrical and chemical processing systems

Standardized Test Scores

GRE: 311 (Q: 160, V: 151, AWA: 3.0)
IELTS: 6.5 (Writing: 7.0, Speaking: 6.0)

Additional Coursework

Quantum Computing, Mathematics for Machine Learning, Multivariate Calculus, Mathematical Thinking, Digital Signal Processing, Theory of Computation, Algorithms (MIT OpenCourseWare, Coursera)

Honors and Awards

- 2nd Place, National Undergraduate Mathematical Olympiad (2016)
- KUET Merit Scholarship (2013, 2014)
- Runner-up, Bangladesh Mathematical Olympiad (2011, 2012)

Languages

Bengali (Native), English (Fluent), French (Basic), Urdu (Basic), Hindi (Basic)